

CSA 0378-DATA STRUCTURES

17. INSERTION SORT

main.c	Run	Output
<pre>1 #include <stdio.h> 2 3 int main() { 4 int n, i, j, key; 5 6 printf("Enter number of elements: "); 7 scanf("%d", &n); 8 9 int arr[n]; 10 11 printf("Enter elements: "); 12 for(i = 0; i < n; i++) 13 scanf("%d", &arr[i]); 14 15 for(i = 1; i < n; i++) { 16 key = arr[i]; 17 j = i - 1; 18 19 while(j >= 0 && arr[j] > key) { 20 arr[j + 1] = arr[j]; 21 j--; 22 } 23 24 arr[j + 1] = key; 25 } 26 27 printf("Sorted Array: "); 28 for(i = 0; i < n; i++) 29 printf("%d ", arr[i]); 30 31 return 0; 32 }</pre>	   Share	Enter number of elements: 4 Enter elements: 20 30 10 50 Sorted Array: 10 20 30 50 === Code Execution Successful ===